Successive multivariate variational mode decomposition

Published: 29 March 2022
Volume 33, pages 917–943, (2022) Cite this article

extracting mode has no or less spectral overlap with the previously obtained modes and the residual signal. Finally, we employ the alternate direction method of the multiplier algorithm (ADMM) to solve it. Compared with other multivariate extending methods whose performances will be degraded if the number of modes is not precisely known, this extension can recursively extract modes and does not need to know the number of modes. Therefore, it achieves better performance on convergence and computation requirements. Moreover, it is more robust to the initial center frequency and possesses the mode-alignment property. We also investigate the relationships between the regularization parameter α and the spectrum property of modes. Some suggestions for selecting proper solution parameters are provided. Finally, we show promising practical decomposition results on a series of simulating and real-life multi-channel data.

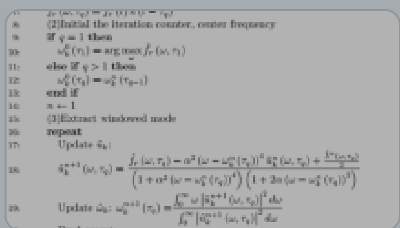
Buy article PDF 39,95 €

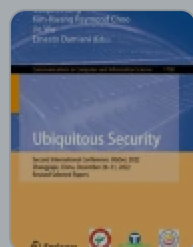
Price includes VAT (Sweden)

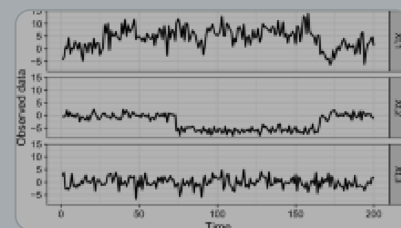
Instant access to the full article PDF.

Institutional subscriptions →

Similar content being viewed by others







References

Bao, C., Hao, H., Li, Z. X., & Zhu, X. (2009). Time-varying system identification using a newly improved HHT algorithm. *Computers & Structures, 87*(23), 1611–1623.

Article Google Scholar

Chen, Q., Xie, L., & Su, H. (2020). Multivariate nonlinear chirp mode decomposition. *Signal Processing, 176,* 107667.

Article Google Scholar

Chen, S., Dong, X., Peng, Z., Zhang, W., & Meng, G. (2017). Nonlinear chirp mode decomposition: A variational method. *IEEE Transactions on Signal Processing, 65*(22), 6024–

Article MathSciNet Google Scholar

Feng, Z., Liang, M., & Chu, F. (2013). Recent advances in time–frequency analysis methods for machinery fault diagnosis: A review with application examples. *Mechanical Systems and Signal Processing, 38*(1), 165–205.

Article Google Scholar

Hao, H., Wang, H. L., & Rehman, N. U. (2017). A joint framework for multivariate signal denoising using multivariate empirical mode decomposition. *Signal Processing, 135,* 263–273.

Article Google Scholar

Article MathSciNet Google Scholar

Jiang, X., Shen, C., Shi, J., & Zhu, Z. (2018). Initial center frequency-guided VMD for fault diagnosis of rotating machines. *Journal of Sound and Vibration, 435*, 36–55.

Article Google Scholar

Jiang, X., Wang, J., Shi, J., Shen, C., Huang, W., & Zhu, Z. (2019). A coarse-to-fine decomposing strategy of VMD for extraction of weak repetitive transients in fault diagnosis of rotating machines. *Mechanical Systems and Signal Processing, 116*, 668–692.

Article Google Scholar

Li, Z., Jiang, Y., Guo, Q., Hu, C., & Peng, Z. (2018). Multi-dimensional variational mode decomposition for bearing-crack detection in wind turbines with large driving-speed variations. *Renewable Energy, 116,* 55–73.

Article Google Scholar

Liu, S., & Yu, K. (2022). Successive multivariate variational mode decomposition based on instantaneous linear mixing model. *Signal Processing, 190,* 108311.

Article Google Scholar

Looney, D., & Mandic, D. P. (2009). Multiscale image fusion using complex extensions of emd. *IEEE Transactions on Signal Processing, 57*(4), 1626–1630.

Rehman, N., & Mandic, D. P. (2010). Multivariate empirical mode decomposition. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences, 466*(2117), 1291–1302.

Article MathSciNet Google Scholar

Rehman, N., Looney, D., Rutkowski, T.M., & Mandic, D.P. (2009). Bivariate EMD-based image fusion. In: 2009 IEEE/SP 15th Workshop on Statistical Signal Processing, pp. 57–60

Rehman, N., Xia, Y., & Mandic, D.P. (2010). Application of multivariate empirical mode decomposition for seizure detection in EEG signals. In: 2010 Annual International Conference of the IEEE Engineering in Medicine and Biology, pp. 1650–1653

Article Google Scholar

Sun, H., Fang, L., & Zhao, F. (2019). A fault feature extraction method for single-channel signal of rotary machinery based on VMD and KICA. *Journal of Vibroengineering, 21*(2), 370–383.

Article Google Scholar

Wu, Z., & Huang, N. E. (2009). Ensemble empirical mode decomposition: A noise-assisted data analysis method. *Advances in Adaptive Data Analysis, 01*(01), 1–41.

Article Google Scholar

Acknowledgements

The helpful and constructive comments from the referees have led to the improvements of this paper; the authors gratefully acknowledge this assistance.

Author information

Authors and Affiliations

Department of Astronautic Science and Mechanics, Harbin Institute of Technology, No.92 West Dazhi Street, Harbin, 150001, China

Shuaishuai Liu & Kaiping Yu

Corresponding author

Appendix A

In this appendix, we show that Eq. (28) is tenable around the center frequency $\omega _k$ and has the same effective bandwidth as Eq. (6). The filter in Eq. (20) can be expressed as:

$$\begin{aligned} {H_k}\left({\omega ,{\omega _k},\left\{ {{\omega _i}} \right\} _{i = 1}^{k - 1}} \right) = \frac{1}{{1 + 2\alpha {{\left({\omega - {\omega _k}} \right) }^2} + \sum \limits _{i = 1}^{k - 1} {\frac{1}{{{\alpha ^2}{{\left({\omega - {\omega _i}} \right) }^4}}}} }} \end{aligned}$$

(A.1)

Assume that the algorithm of SMVMD is converged to the true modes and also, the modes are spectrally distinct. When ω is close to $\omega _k$, then $\left| {\omega \mathrm{{ - }}{\omega _i}} \right| \ge \rho > 0$ and the following relation is given:

(B.1)

where $\varvec{a}_k$ with the same size as $\varvec{\hat{u}}_k(\omega)$ is the k-th receiver steering vector and $\hat{s}_k(\omega)$ denotes the corresponding source. The following operation can be conducted:

$$\begin{aligned} \int _{ - \infty }^{ + \infty } {\varvec{\hat{u}}_k(\omega)\cdot \hat{u}_{k,c}^{H}(\omega) \mathrm{d} \omega }&=\int _{ - \infty }^{ + \infty } {\varvec{a}_k \hat{s}_k(\omega)\cdot a_{k,c} \hat{s}_k^{H}(\omega) \mathrm{d} \omega }\\&= \varvec{a}_k\int _{ - \infty }^{ + \infty } {a_{k,c} \left| \hat{s}_k(\omega) \right| ^2 \mathrm{d} \omega }\\ \end{aligned}$$

(B.2)

\omega }}{\int _{ - \infty }^{ + \infty } {{{{\hat{u}}}}}_{k,1}(\omega)\cdot {{{\hat{u}}}}_{k,c}^{H}(\omega) \mathrm{d} \omega } \end{aligned} \end{aligned}$$

(B.4)

Then \({{{\hat{s}}}}_k{(\omega)}\) can be recovered based on Eq. (B.1):

$$\begin{aligned} \begin{aligned} {{{\hat{s}}}}_k{(\omega)} = \frac{{{\varvec{a}}}_k^H{{{\hat{\varvec{u}}}}}_k(\omega)}{{{\varvec{a}}}_k^H{{\varvec{a}}}_k} \end{aligned} \end{aligned}$$

(B.5)

Rights and permissions

Keywords

Multivariate signal processing
Variational mode decomposition
Successive extraction
Compact bandwidth
Filterbank spectrum property